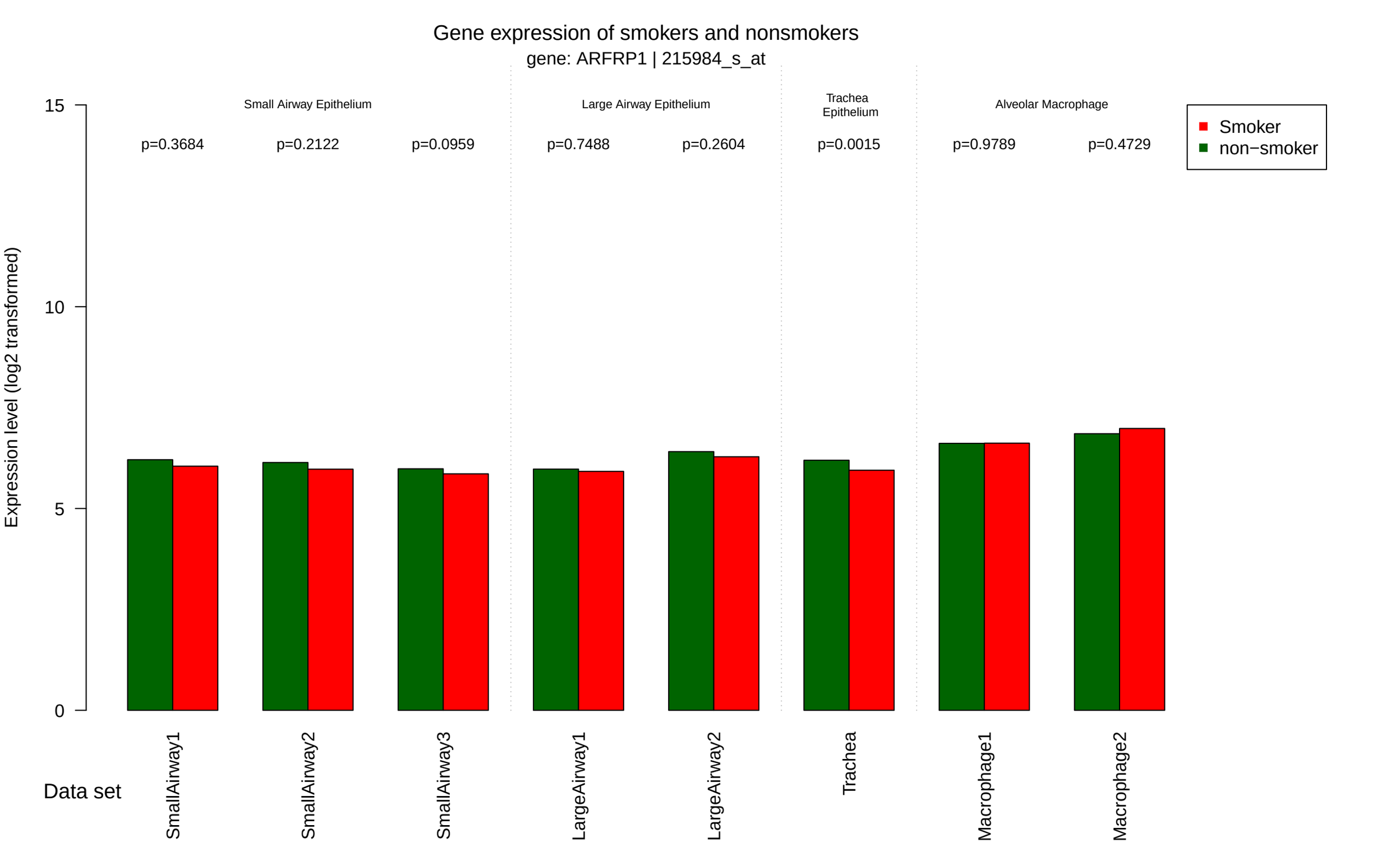
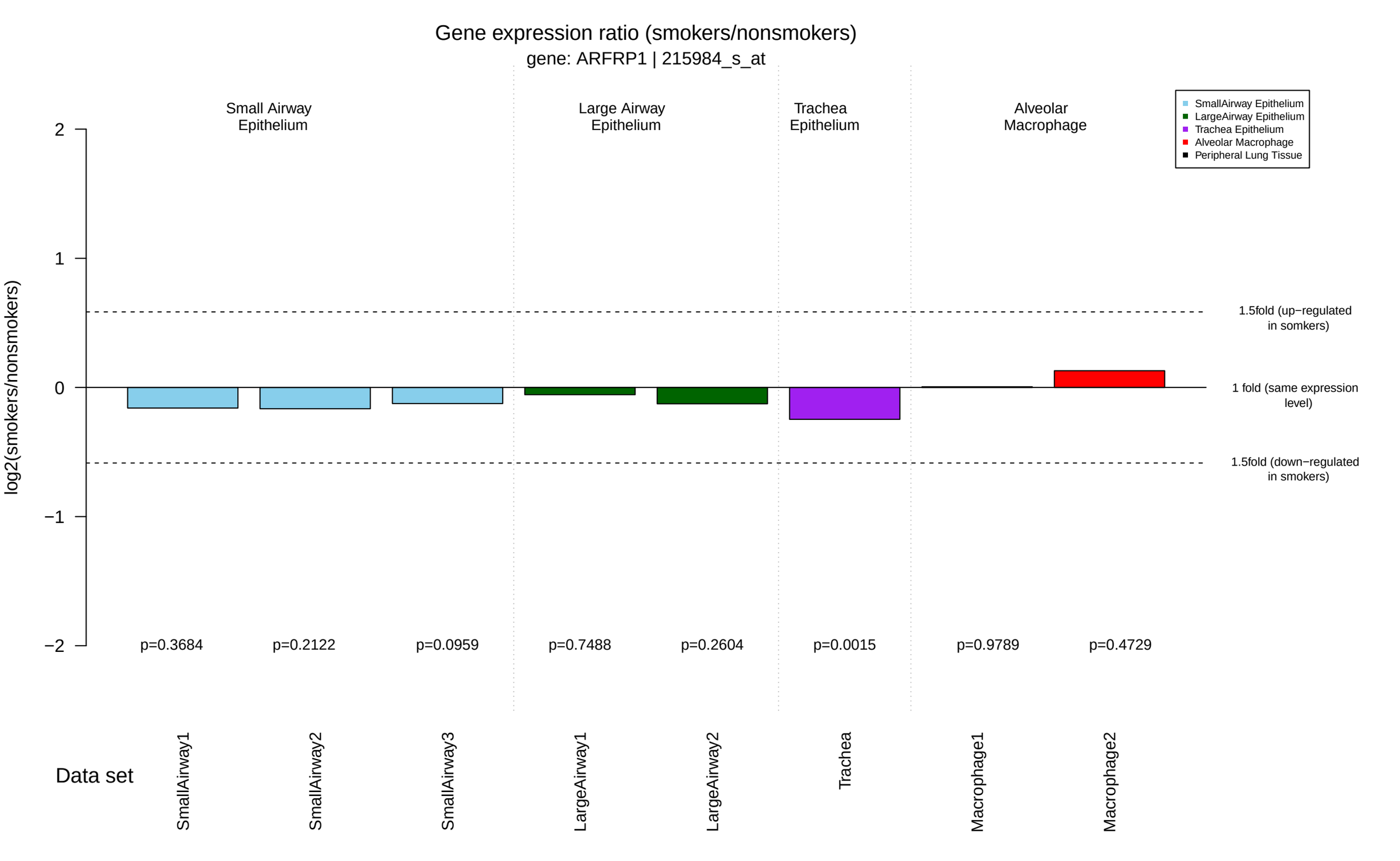
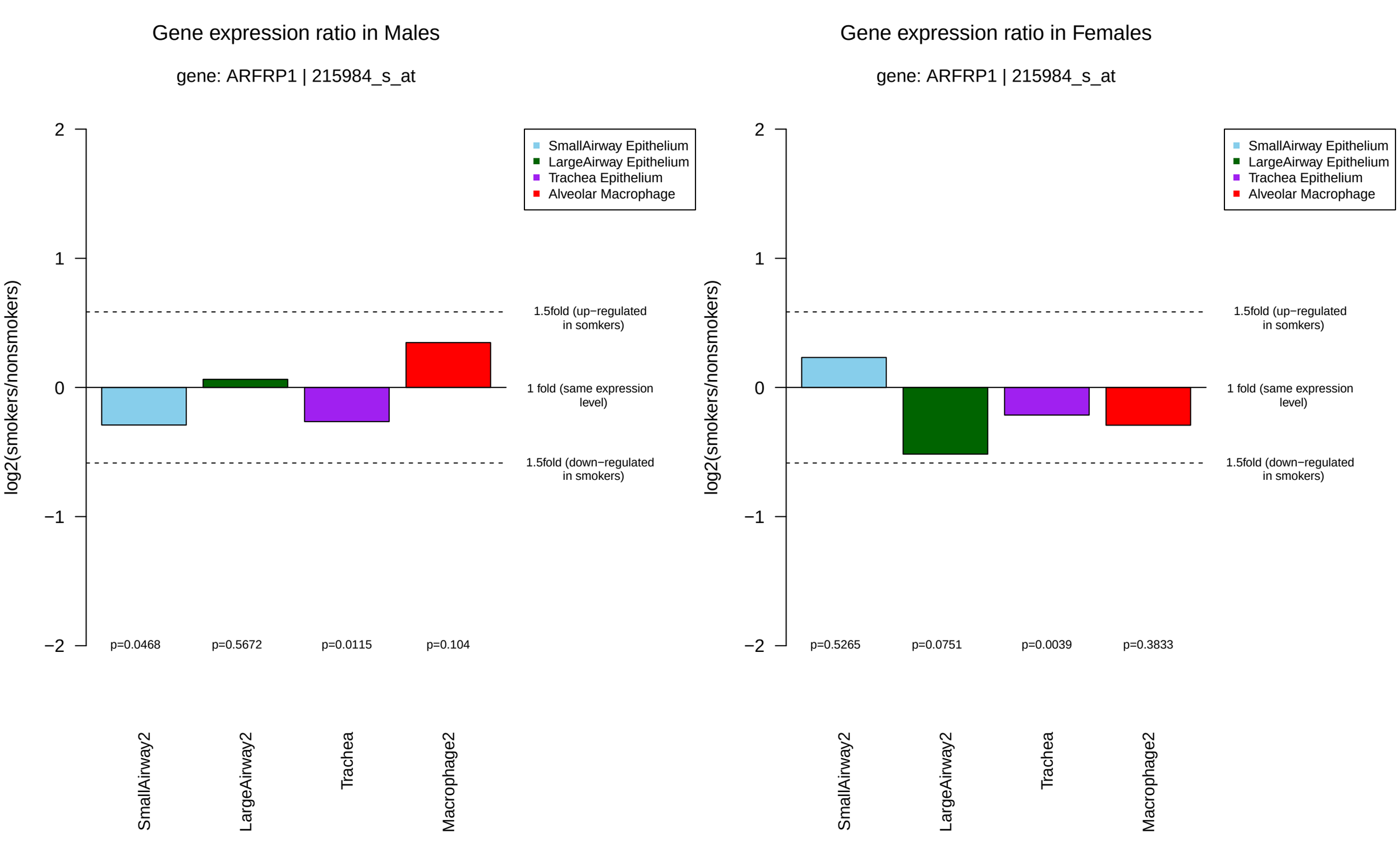
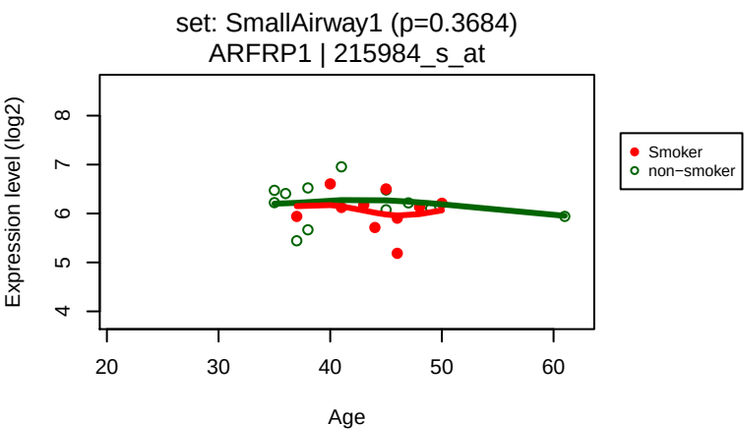
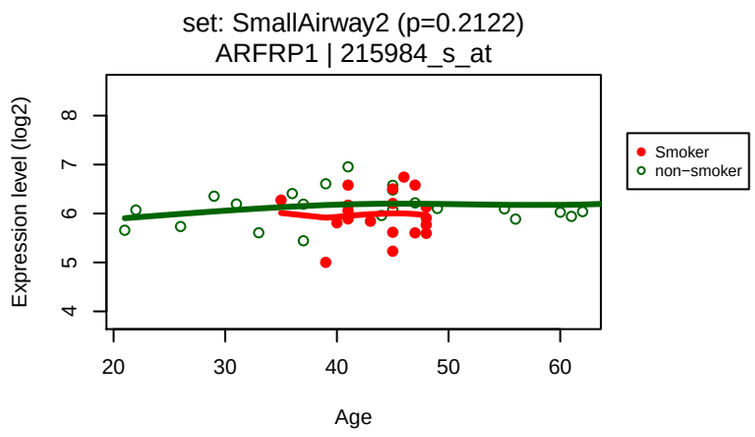
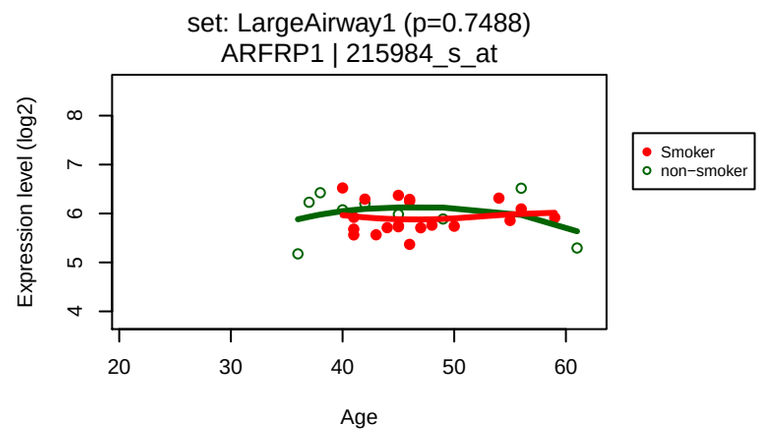

Gene expression of smokers and nonsmokers
gene: ARFRP1 | 215984_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3684
p=0.2122
p=0.0959
p=0.7488
p=0.2604
p=0.0015
p=0.9789
p=0.4729
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

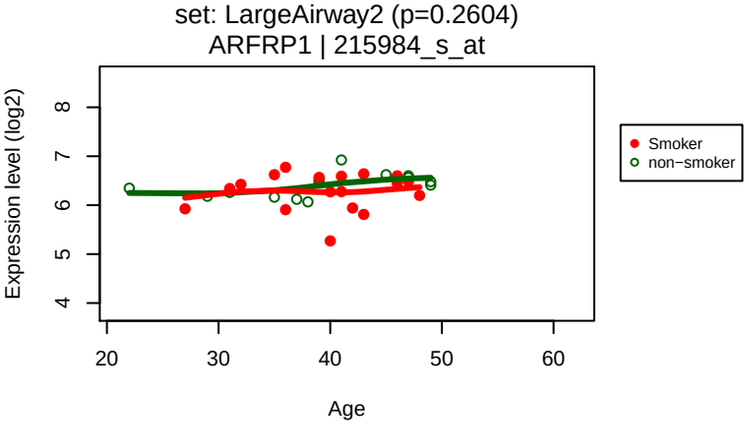

Gene expression ratio (smokers/nonsmokers)
gene: ARFRP1 | 215984_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3684
p=0.2122
p=0.0959
p=0.7488
p=0.2604
p=0.0015
p=0.9789
p=0.4729
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

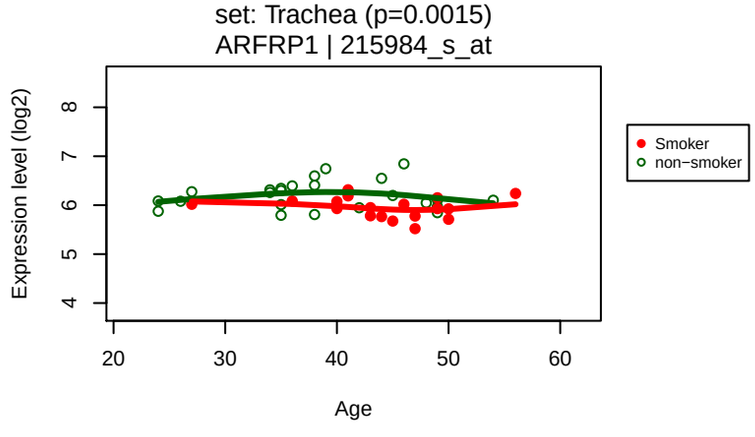

Gene expression ratio in Males
gene: ARFRP1 | 215984_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0468
p=0.5672
p=0.0115
p=0.104
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARFRP1 | 215984_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5265
p=0.0751
p=0.0039
p=0.3833
SmallAirway2
LargeAirway2
Trachea
Macrophage2

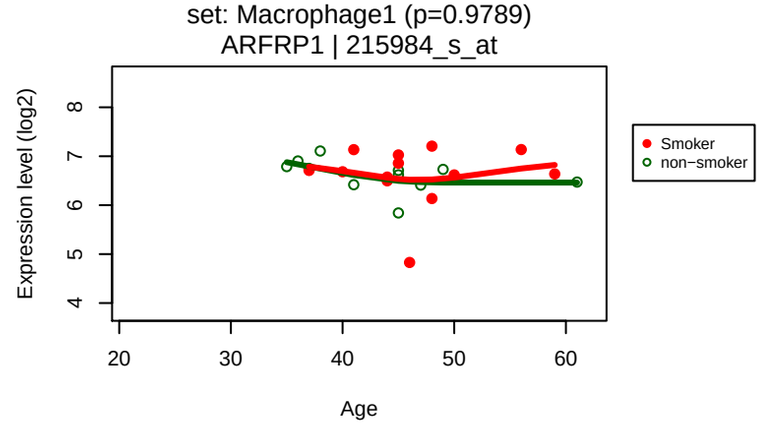

set: SmallAirway1 (p=0.3684)
ARFRP1 | 215984_s_at
Expression level (log2)
Age
Smoker
non−smoker

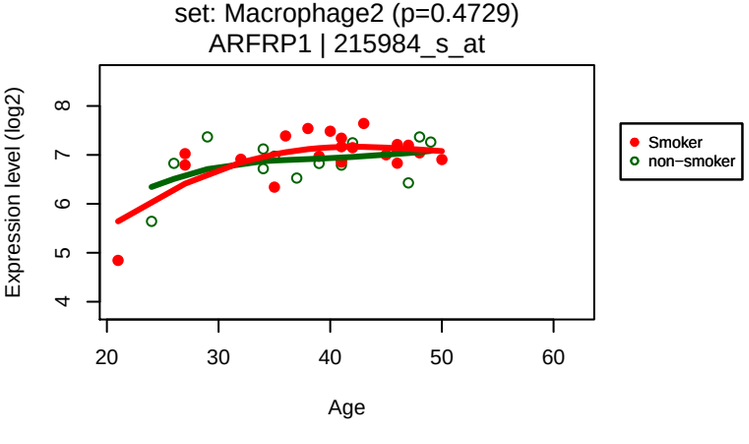

set: SmallAirway2 (p=0.2122)
ARFRP1 | 215984_s_at
Expression level (log2)
Age
Smoker
non−smoker

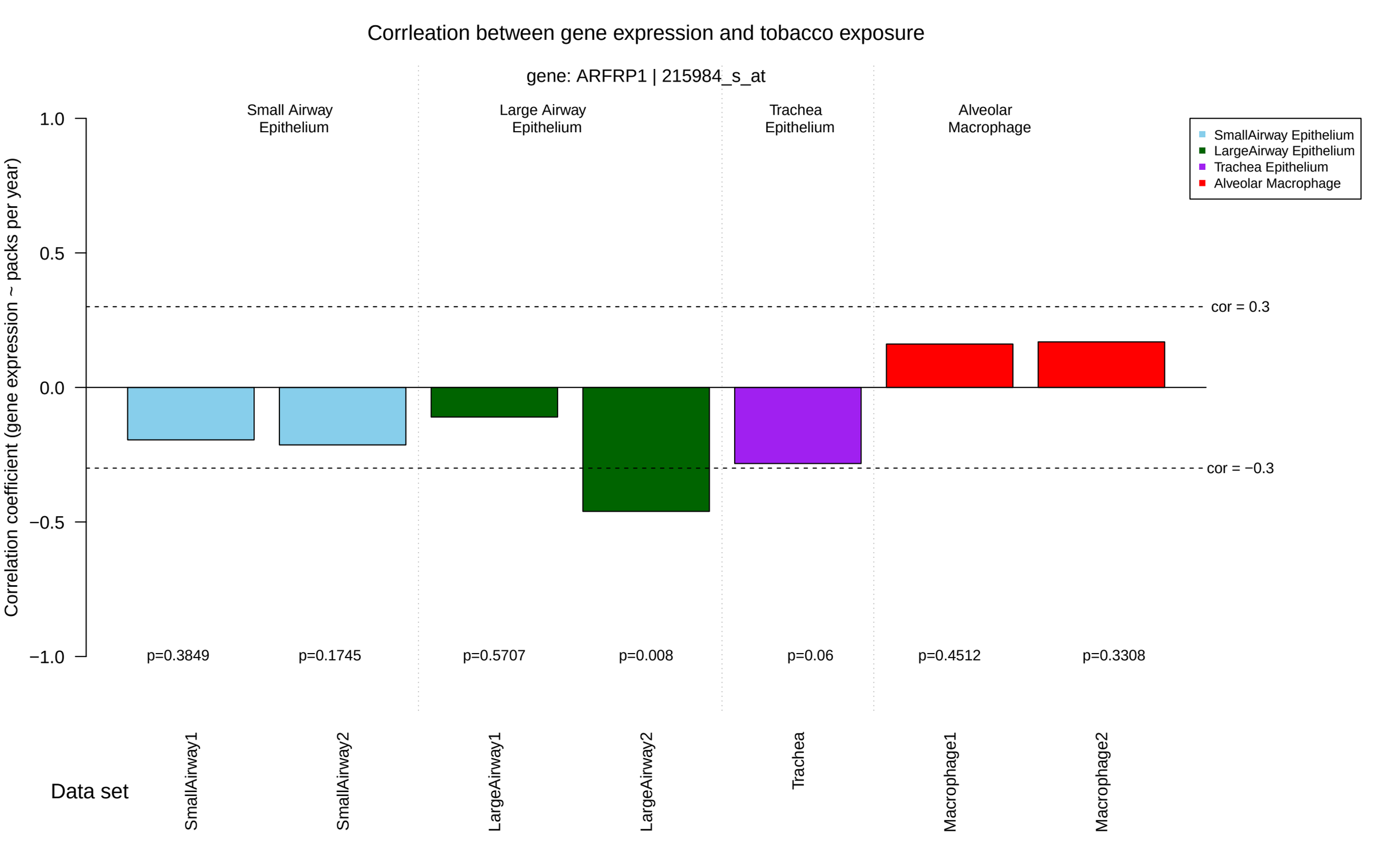

set: LargeAirway1 (p=0.7488)
ARFRP1 | 215984_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2604)
ARFRP1 | 215984_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0015)
ARFRP1 | 215984_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9789)
ARFRP1 | 215984_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4729)
ARFRP1 | 215984_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARFRP1 | 215984_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3849
p=0.1745
p=0.5707
p=0.008
p=0.06
p=0.4512
p=0.3308
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2